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Name (initial) S. J

Midterm Examination
CHE 230
Molecular Systems in Chemical Engineering

UIC Chemical Engineering Department

03/02/2026

Time: 70 minutes

Honor Code: As a commitment to the personal, academic, and professional standards upheld by the Honors College, as a student, I hereby pledge to respect, preserve, and promote the Honor Code of Conduct during my entire tenure at the Honors College. I agree to uphold high ethical principles in all personal, academic, and professional endeavors related to the Honors College and beyond, and to always be mindful of being a representative of the Honors College and the UIC community

Signature [signature]

Date 03/02/2026

Section 1: Fundamental to Molecular Systems (30 points) 15

1. **(5 points)** What is the **difference** between a physical and a chemical bond? Give **one** example of each.

Physical properties are due to atomic arrangements which can cause things like brittleness or ductility.

Chemical properties are due to the atoms themselves like bonding, number of protons/electrons which affects how they interact with other molecules. One example would be a materials ability to corrode.

3

2. **(5 points)** Graphite and diamond are chemically identical yet have different physical properties. Please briefly explain why.

Graphite and diamond both share the same chemical formula but differ in atomic structure. This difference in structure causes diamond to be harder and graphite more brittle.

4

3. **(5 points)** Briefly cite the main differences between ionic, covalent, and metallic bonding.

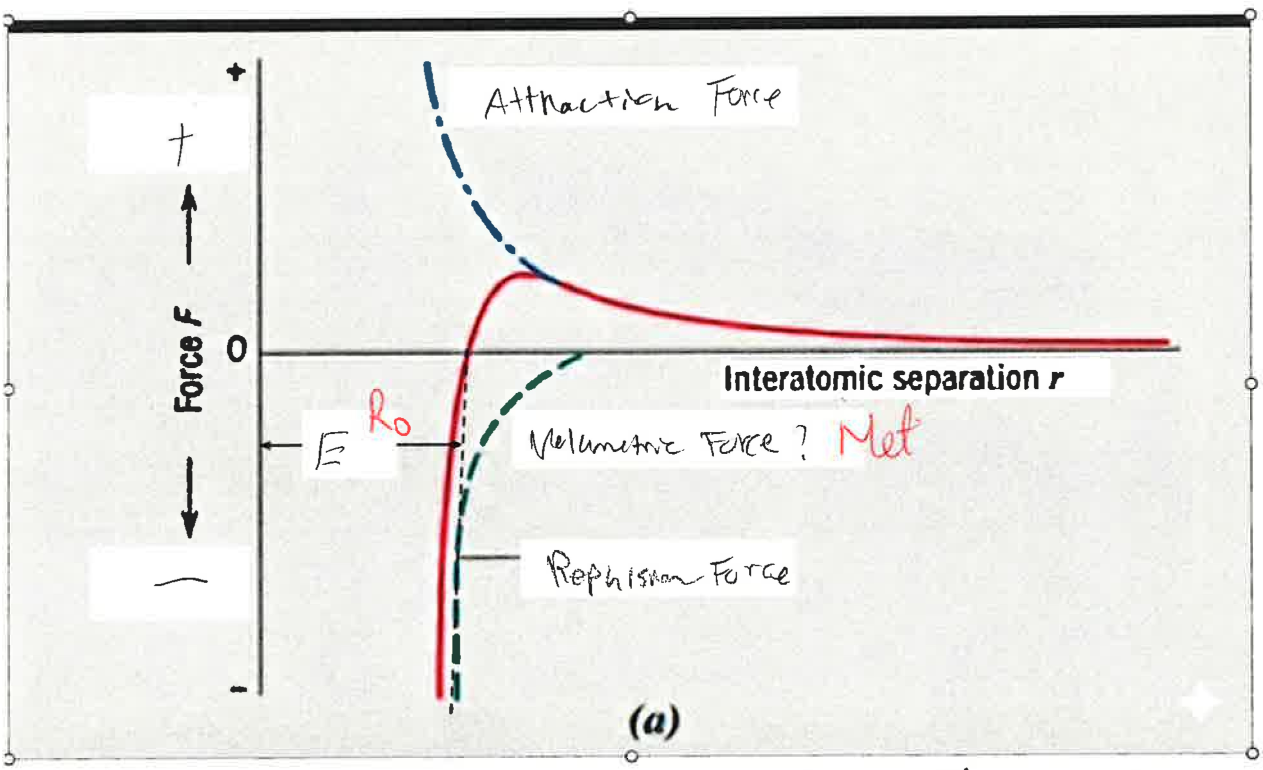
Ionic is the transer of electrons.
Covalent is the sharing between electrons.
Metallic is positive ions with an electron cloud of negative charges.

3

4. **(6 points)** The graph provided illustrates the forces acting between two atoms as a function of their interatomic separation, r.

Fill in the blank boxes on the diagram with the correct terms.

2.



$F = kg \cdot \frac{m^2}{s}$ so $F \times r = kg \frac{m^3}{s}$

5. **(9 points)** The atomic radii of Mg^{2+} and O^{2-} ions are 0.072 nm and 0.140 nm, respectively.

(a) Calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another)

(b) What is the force of repulsion at this same separation distance?

Given Constants:

- Magnitude of electrical charge for an electron (e): 1.602×10^{-19} C
- Permittivity of a vacuum ϵ_o: 8.854×10^{-12} F/m (or $C^2/(N.m^2)$)

(a) $F = \frac{Ze^2}{\varepsilon_o (\frac{F}{m} \text{ or } \frac{C^2}{N \cdot m^2}) A^2} = \frac{(1.602 \times 10^{-19})^2}{8.854 \times 10^{-12} \frac{C^2}{N m^2} (0.072 \times 10^{-9})(0.140 \times 10^{-9})}$

$= 2.89 \times 10^{-7} N$ ✗ B

$Z = 2$

(b) Force of repulsion is $-2.89 \times 10^{-7} N$ ✗

Section 2: Crystal Solid (25 points) 17.

1. (**5 points**) Calculate the radius of an iridium atom, given that Ir has an FCC crystal structure, a density of 22.4 g/cm^3, and an atomic weight of 192.2 g/mol(N_A = 6.022 X 10^{23} atoms/mol).

4 atoms

find a?

$4 \text{ atoms} \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ atoms}} \times \frac{192.2 \text{ g}}{1 \text{ mol}} \times \sqrt[3]{\frac{1 \text{ cm}^3}{22.4 \text{ g}}} =$

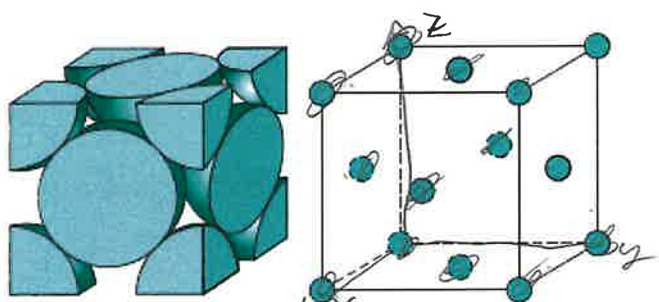
3.85×10^{-8}

$= 0.385$ nm

X 1.

2. (**10 points**) List the point coordinates for all atoms that are associated with the FCC unit cell (Figure 2)

(0, 0, 0) (1/2, 1/2, 0) (0, 1/2, 1/2)
(1, 0, 0) (0, 0, 1) (0, 1, 1)
(1/2, 1/2, 0) (1, 1, 0) (1/2, 1, 1/2)
(1, 0, 1) (1/2, 1/2, 1)
(1/2, 1/2, 1/2) (1, 1, 1)



(a) (b)

Figure 2 – FCC structure

✓ 10

3. **(5 points)** What is the unit cell number, coordination number, Edge Length and APF of a Body Centered Cubic (Show calculation of APF).

4 atoms

edge length = $2\sqrt{2}r$

$4r = a\sqrt{2}$

$APF = \frac{V_{atoms}}{V_{cells}} = \frac{4 \times \frac{4}{3}\pi r^3}{16\sqrt{2}r^3} = 0.236$

$V_{atoms} = \frac{4}{3}\pi r^3$

$V_{cells} = a^3 = (2\sqrt{2}r)^3$

X 1

4. **(5 points)** Match each definition or description below with the most appropriate term from the word bank. Each term may be used once, more than once, or not at all.

5

Word Bank

Grain Boundary	**Substitutional impurity**	**Single ~~Crystal~~**	**Vacancy Defect**
Interstitial Impurity	**~~Coordination~~ Number**	**Edge ~~Length~~**	**Unit Cell**

Coordination # ____ The number of nearest neighbor or touching atom. ✓

Grain boundary ____ A two-dimensional planar defect that separates regions of identical crystal structure but differing crystallographic orientations. ✓

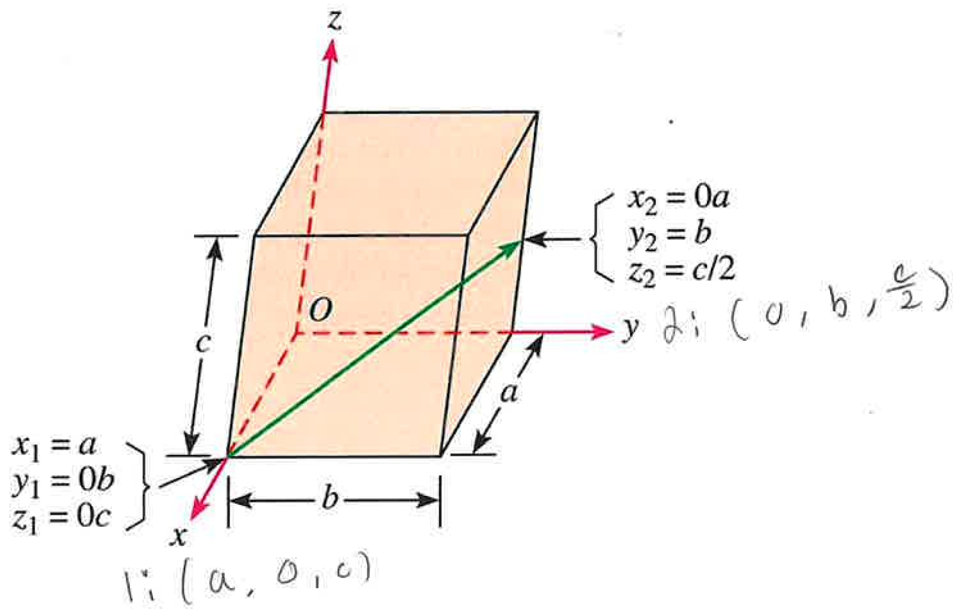
interstitial impurity ____ Foreign atom replaces host atom. ✓

Edge length ____ The physical dimension or length of one side of a unit cell. ✓

Single crystal ____ A crystalline solid in which the perfectly ordered, repeating arrangement of atoms is continuous and unbroken throughout the entire sample. ✓

Extra Credit

(5 points) Determine the indices for the direction shown in the accompanying figure.



Section 3. Material Properties (45 points) 21

Diffusion

1. (**10 points**) For some applications, it is necessary to harden the surface of a steel (or iron–carbon alloy) above that of its interior. One way this may be accomplished is by increasing the surface concentration of carbon in a process termed carburizing; the steel piece is exposed, at an elevated temperature, to an atmosphere rich in a hydrocarbon gas, such as methane (CH_4). Consider one such alloy that initially has a uniform carbon concentration of 0.25 wt% and is to be treated at 950°C (1750°F). If the concentration of carbon at the surface is suddenly brought to and maintained at 1.20 wt%, how long will it take to achieve a carbon content of 0.80 wt% at a position 0.5 mm below the surface? The diffusion coefficient for carbon in iron at this temperature is 1.6 × 10–11 m^2 /s; assume that the steel piece is semi-infinite.

Error Function is on the last page

t ?

$C_0 = 0.25$

$T = 950\,°C$

$C_s = 1.20\,wt\%$

$C_x = 0.80\,wt\%$

$x = 0.5\,mm$

$D = 1.6 \times 10^{-11}\,m^2/s$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$D = D_0 \exp\left(\frac{-Q}{RT}\right)$$?

$$J = -D\frac{dc}{dx}$$

$$\frac{0.80 - 0.25}{1.20 - 0.25} = 1 - erf\left(\frac{0.0005\,m}{2\sqrt{1.6\times10^{-11}\,t}}\right)$$

$$0.421 = erf\left(\frac{x}{2\sqrt{D}\,t}\right)$$

$= 0.4475$

3.925

$$\frac{x}{2\sqrt{D}\,t} = 0.4475$$

$t = 1396.64\,S$

About a day

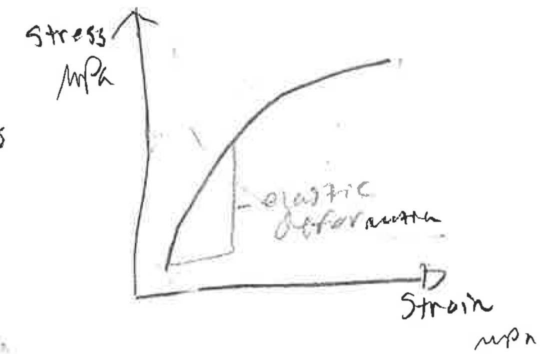
6

2. (4 points) **Distinguish** between elastic and plastic deformations, both by definition, and in terms of behavior on a stress-strain plot. **Draw** a sketch to illustrate your answer.

Elastic deformation is an materials ability to stretch using %EL.

Plastic deformation is the materials ability to undergo strain before fracture point.

The plastic deformations have less ability to undergo strain then elastic deformations.



2

3. (3 points) **Highlight** the differences between true stress and engineering stress.

True stress is an applied force whereas engineering stress is caused by mechanical force.

X

4. (**8 points**) Of the metals listed in the Table below,

(a) Which will experience the greatest percent reduction in area? Why? B low Elasticity
(b) Which is the strongest? Why? D, highest yield and tensile strength
(c) Which is the stiffest? Why? B, low electric modulus
(d) Which is the hardest? Why? D, highest tensile strength

6

Material	Yield Strength (MPa)	Tensile Strength (MPa)	Strain at Fracture	Fracture Strength (MPa)	Elastic Modulus (GPa)
A	310	340	0.23	265	210
B	100	120	0.40	105	150
C	415	550	0.15	500	310
D	700	850	0.14	720	210
E	Fractures before yielding			650	350

5. **(10 points)** Consider a cylindrical titanium wire 3.0 mm (0.12 in.) in diameter and 2.5 x 10^4 mm (1000 in.) long. When a load of 500 N (112 lbf) is applied, the wire experiences an elongation of 16.5 mm. Assuming that the deformation is totally elastic, calculate the Young's Modulus of the titanium in GPa.

$$\Delta L = \frac{F \cdot L_0}{A_0 E} \qquad A = \pi \left(\frac{D}{2}\right)^2$$

$$0.0165\ m = \frac{500N \cdot 0.003m}{\pi \left(\frac{2.5 \times 10}{2}\right)^2 \cdot E}$$

$$E = 0.185\ GPa$$

3

X

b. A single crystal of some metal has a τ_{crss} of 30.7 MPa and is exposed to a tensile stress of 61 MPa.

(2 points) (i) Will yielding occur when $\phi = 56°$ and $\lambda = 32°$.

(i) $\tau_r = 61 \cos 56 \cos 32$
$= 28.9$ MPa

NO

(2 points) (ii) If not, what stress is necessary?

$$30.7 = \sigma \cos 56 \cos 32$$

$$\sigma = 64.74\ MPa$$

✓ 4

6. **(6 points)** A relatively large plate of a glass is subjected to a tensile stress of 40 MPa. If the specific surface energy and modulus of elasticity for this glass are 0.3 J/m^2 and 69 GPa respectively, determine the maximum length of a surface flaw that is possible without fracture.

$\tau_r = \sigma \cos\phi (\cos\lambda)$

$40 MPa$

$\sigma = 40 MPa$

$E = 0.3 \frac{J}{m^2}$

$\Delta L = \frac{40 MPa}{(0.3)(69 GPa)}$

$= 1.39 mm$

$\Delta L ?$

$\Delta L = \frac{F \cdot L_0}{A \cdot E}$

X

Appendix G: Tables of Error Function

X	ERF	X	ERF	X	ERF	X	ERF
.01	.011283415556	.46	.484655390002	.91	.801882825766	1.36	.945561436561
.02	.022564574692	.47	.493745050886	.92	.806767721548	1.37	.947312398035
.03	.033841222342	.48	.502749670695	.93	.811563558585	1.38	.949016035256
.04	.045111106145	.49	.511668261189	.94	.816271018976	1.39	.950673295805
.05	.056371977797	.50	.520499877813	.95	.820890807273	1.40	.952285119763
.06	.067621594393	.51	.529243619841	.96	.825423649644	1.41	.953852439360
.07	.078857719771	.52	.537898630479	.97	.829870293036	1.42	.955376178641
.08	.090078125841	.53	.546464096935	.98	.834231504340	1.43	.956857253145
.09	.101280593915	.54	.554939250456	.99	.838508069555	1.44	.958296569601
.10	.112462916018	.55	.563323366325	1.00	.842700792950	1.45	.959695025637
.11	.123622896199	.56	.571615763824	1.01	.846810496228	1.46	.961053509513
.12	.134758351820	.57	.579815806164	1.02	.850838017701	1.47	.962372899854
.13	.145867114836	.58	.587922900382	1.03	.854784211454	1.48	.963654065414
.14	.156947033063	.59	.595936497198	1.04	.858649946527	1.49	.964897864843
.15	.167995971427	.60	.603856090848	1.05	.862436106090	1.50	.966105146475
.16	.17901181319	.61	.611681218876	1.06	.866143586635	1.51	.967276748129
.17	.189992461202	.62	.619411461899	1.07	.869773297164	1.52	.968413496920
.18	.200935839019	.63	.627046443338	1.08	.873326158388	1.53	.969516209093
.19	.211839892158	.64	.634585829122	1.09	.876803101938	1.54	.970585689861
.20	.222702589210	.65	.642029327356	1.10	.880205069574	1.55	.971622733262
.21	.233521922982	.66	.649376687963	1.11	.883533012415	1.56	.972628122027
.22	.244295911599	.67	.656627702300	1.12	.886787890165	1.57	.973602627462
.23	.255022599592	.68	.663782202741	1.13	.889970670363	1.58	.974547009343
.24	.265700058954	.69	.670840062235	1.14	.893082327630	1.59	.975462015822
.25	.276326390168	.70	.677801193837	1.15	.896123842937	1.60	.976348383345
.26	.286899723216	.71	.684665550217	1.16	.899096202880	1.61	.977206836583
.27	.297418218547	.72	.691433123139	1.17	.902000398966	1.62	.978038088373
.28	.307880068029	.73	.698103942917	1.18	.904837426915	1.63	.978842839674
.29	.318283495861	.74	.704678077855	1.19	.907608285972	1.64	.979621779524
.30	.328626759459	.75	.711155633654	1.20	.910313978230	1.65	.980375585023
.31	.338908150311	.76	.717536752806	1.21	.912955507973	1.66	.981104921311
.32	.349125994796	.77	.723821613965	1.22	.915533881027	1.67	.981810441565
.33	.359278654974	.78	.730010431299	1.23	.918050104127	1.68	.982492787002
.34	.369364529345	.79	.736103453821	1.24	.920505184299	1.69	.983152586895
.35	.379382053562	.80	.742100964708	1.25	.922900128256	1.70	.983790458591
.36	.389329701129	.81	.748003280598	1.26	.925235941810	1.71	.984407007545
.37	.399205984043	.82	.753810750875	1.27	.927513629295	1.72	.985002827359
.38	.409009453420	.83	.759523756938	1.28	.929734193014	1.73	.985578499828
.39	.418738700070	.84	.765142711455	1.29	.931898632689	1.74	.986134594997
.40	.428392355047	.85	.770668057608	1.30	.934007944941	1.75	.986671671219
.41	.437969090155	.86	.776100268324	1.31	.936063122773	1.76	.987190275231
.42	.447467618426	.87	.781439845491	1.32	.938065155079	1.77	.987690942224
.43	.456886694550	.88	.786687319174	1.33	.940015026158	1.78	.988174195930
.44	.466225115278	.89	.791843246814	1.34	.941913715258	1.79	.988640548706
.45	.475481719787	.90	.796908212423	1.35	.943762196123	1.80	.989090501636